THE CONTEXT: DEEP LEARNING, AI AND MACHINE LEARNING

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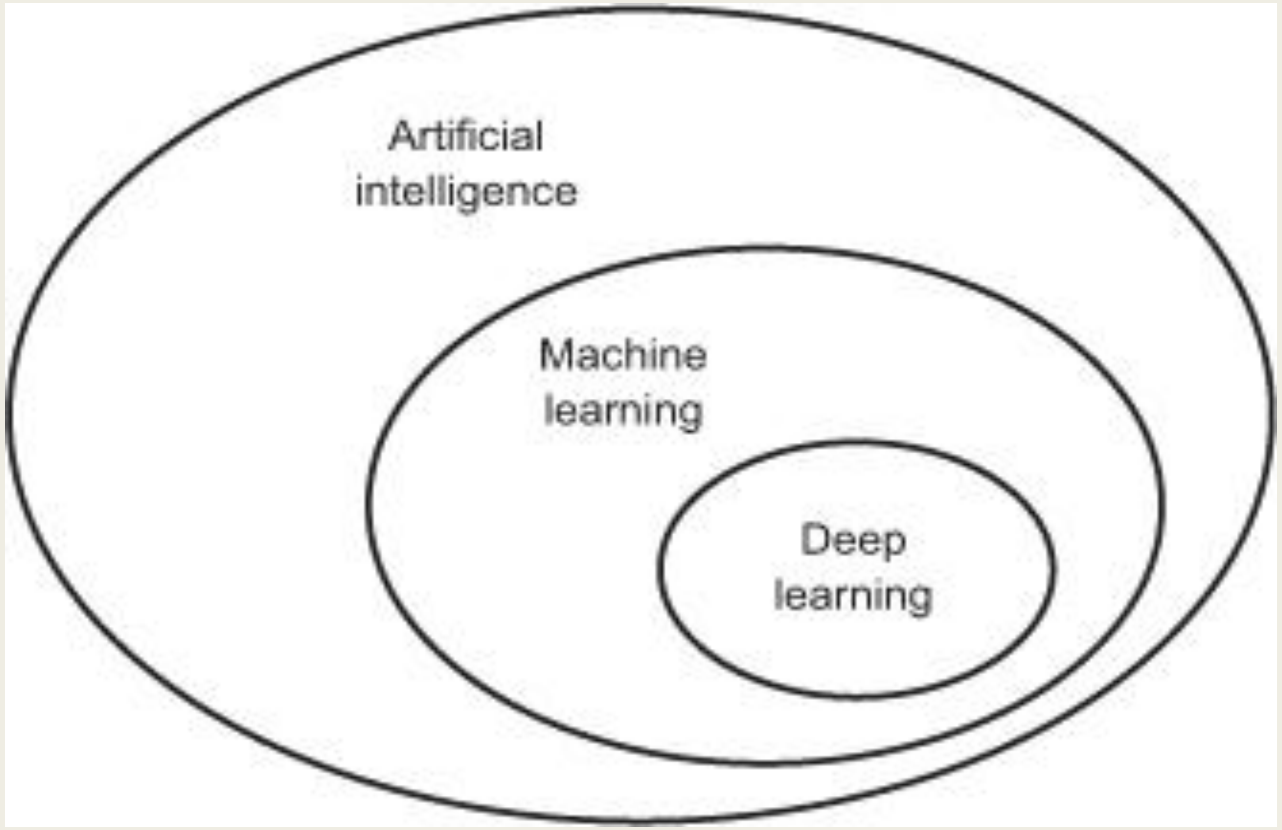
Foreword

- The Context: AI, Machine Learning & Deep Learning
- Building and Implementing Deep Learning Models
- Python & Keras
- Configuration and Model Parameters: Activation Functions, Loss Functions, Optimization, ...etc
- Gradient Descent and Backpropagation
- Examples of Applying Deep Learning Models to Solve Problems in Different Areas – a repeating task (Seeing is Believing, Doing is Learning)

Machine Learning is Subfield of AI



It is not that new – Revival Due to Advances in Technology

Techniques	Year
Neural networks	1943
Backpropagation	1962?
Convolutional Neural Networks	1979
Recurrent Neural Networks	1980
Long-Short-Term-Memory	1997

AI, the Parent Field

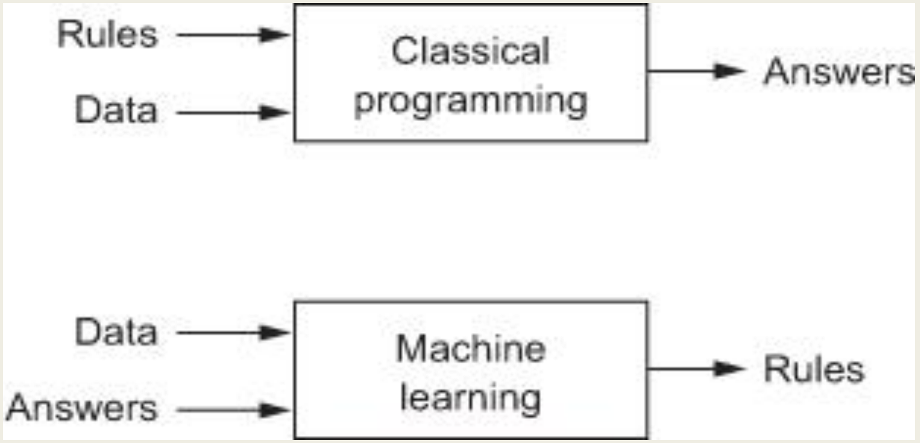
- Born in the 1950's as an attempt to answer the question: Can Machines Think?
- One popular definition of AI : *the effort to automate intellectual tasks normally performed by humans*.
- Traditional symbolic AI is based on the assumption that computers (machines) can achieve human level intelligence by having programmers (developers) hand-craft a large set of rules that manipulate knowledge and conditions (examples: playing chess- Expert Systems- Syntax and Semantic Rules for NLP, Sentiment Analysis)
- Accordingly, symbolic AI systems are rule based and rely on human beings to build rule-bases or knowledge bases
- Examples are chess playing programs and expert systems

Machine Learning?

- In 1843, Ada Lovelace commented on the intelligence of the Mechanical Engine (Computer) invented by Babbage (the first Mechanical Computer) : "The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform.... Its province is to assist us in making available what we're already acquainted with."
- Turing, 1950 commented on Ada objection and concluded that general purpose machines (computers) are capable of learning and originality.
- Later, machine learning emerged as an answer to the same Ada objection: could a computer go beyond "what we know how to order it to perform" and learn on its own how to perform a specified task?

Machine Learning Systems Discover Rules From Data

- Machine-learning systems are *trained* rather than explicitly programmed.
- They are presented with many examples and patterns relevant to the problem, and they discover some structures in these examples that makes the system capable of generating rules that automate the solution of the problem and the performance of a general task. For example image recognition.



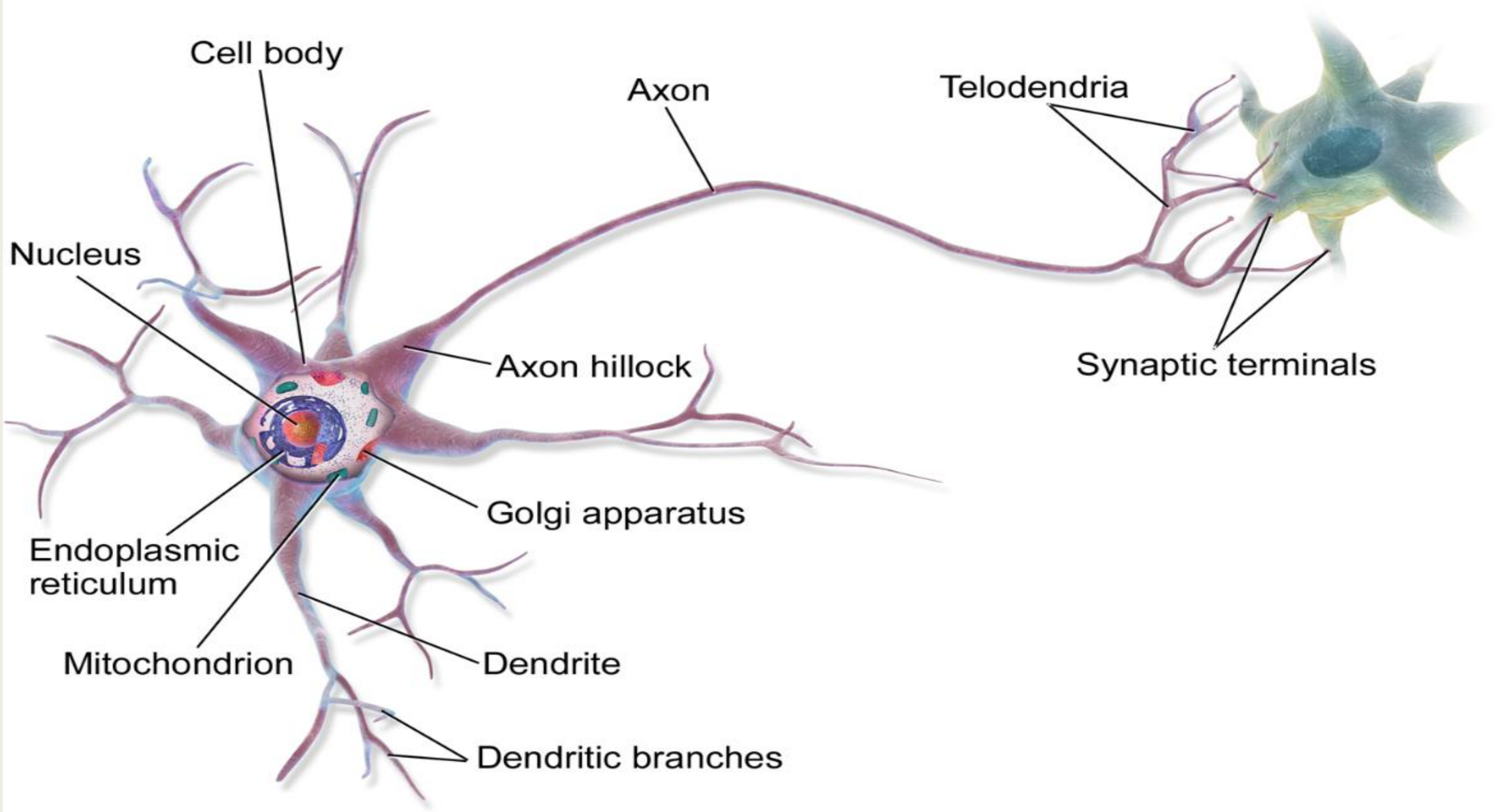
What is Machine Learning?

- Machine Learning is: “searching for useful representations of some input data, within a predefined space of possibilities, using guidance from a feedback signal”.
- This simple idea makes it possible to solve a broad range of intellectual tasks, from speech recognition to autonomous car driving.

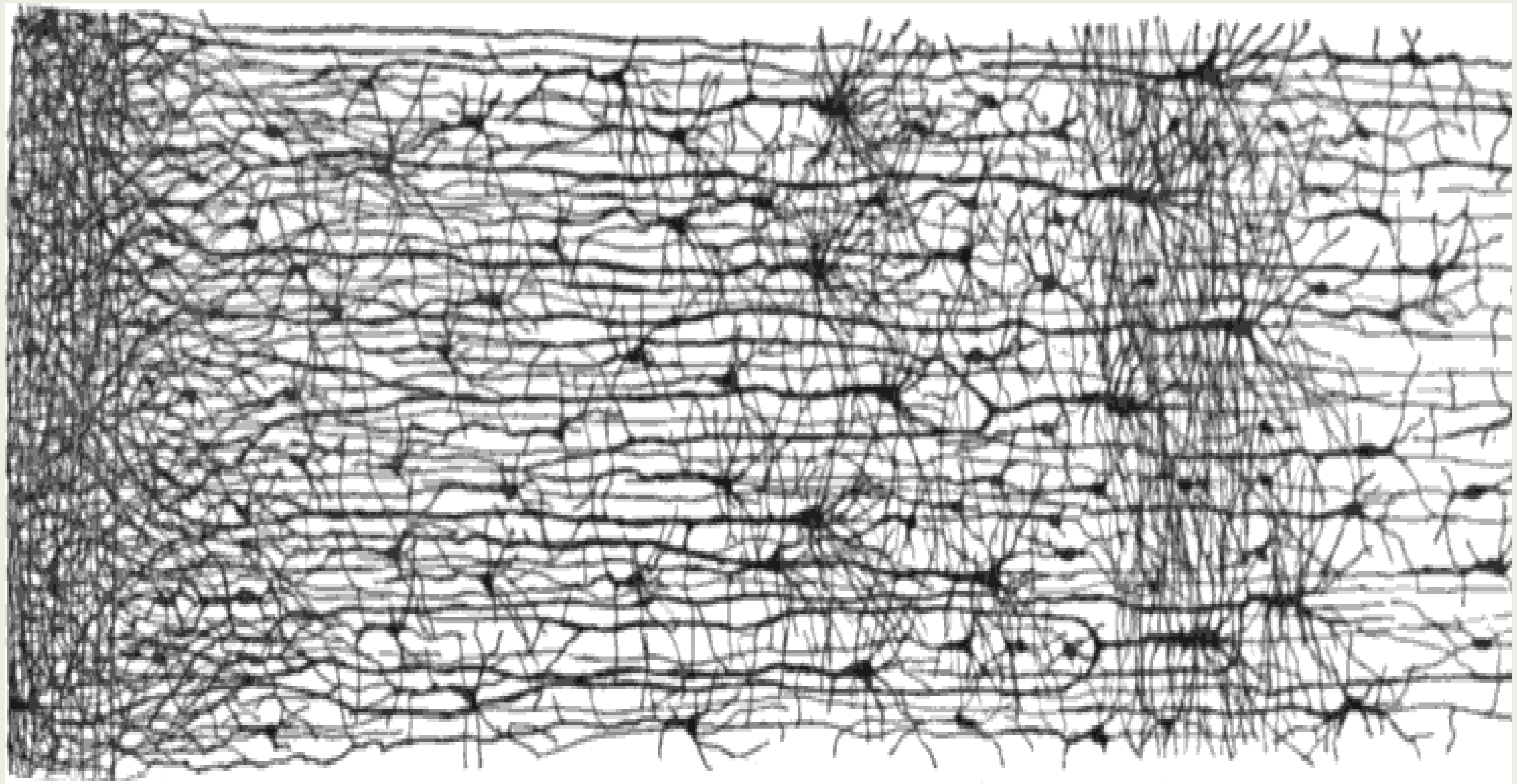
Deep Learning

- a new approach on learning representations from data that puts an emphasis on learning successive *layers* of increasingly meaningful representations.
- Deep learning is a variation and subfield of machine learning where the emphasis is on learning representations using multi-layer neural networks
- The word “deep” here refers to the number of hidden (intermediate) layers in the neural network – usually more than two hidden layers indicates a deep network
- Another difference from traditional machine learning is that features are extracted by the model, while in machine learning feature extraction is made through some manual tasks

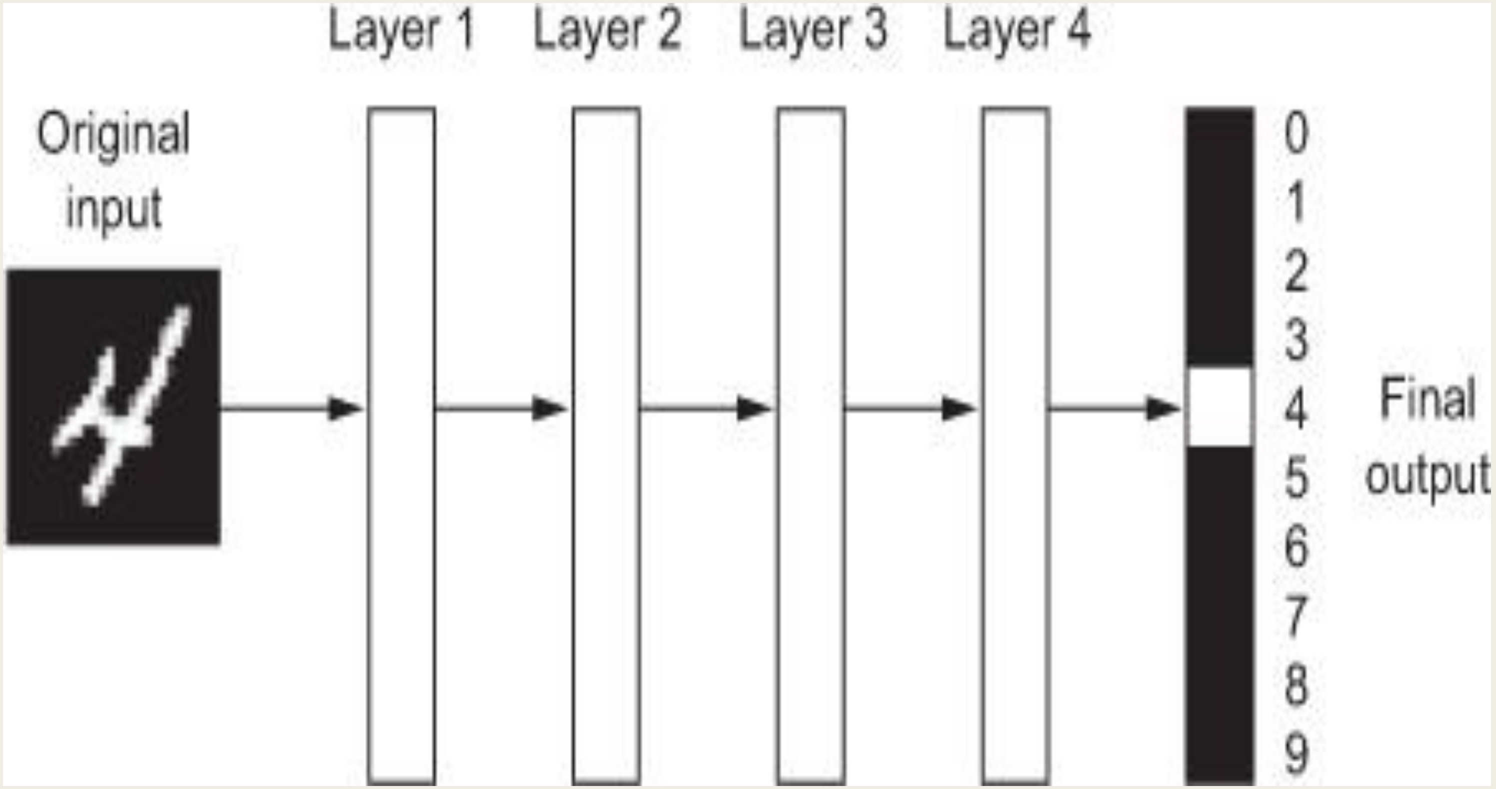
Neural Networks: Lessons From Biology
Neurons



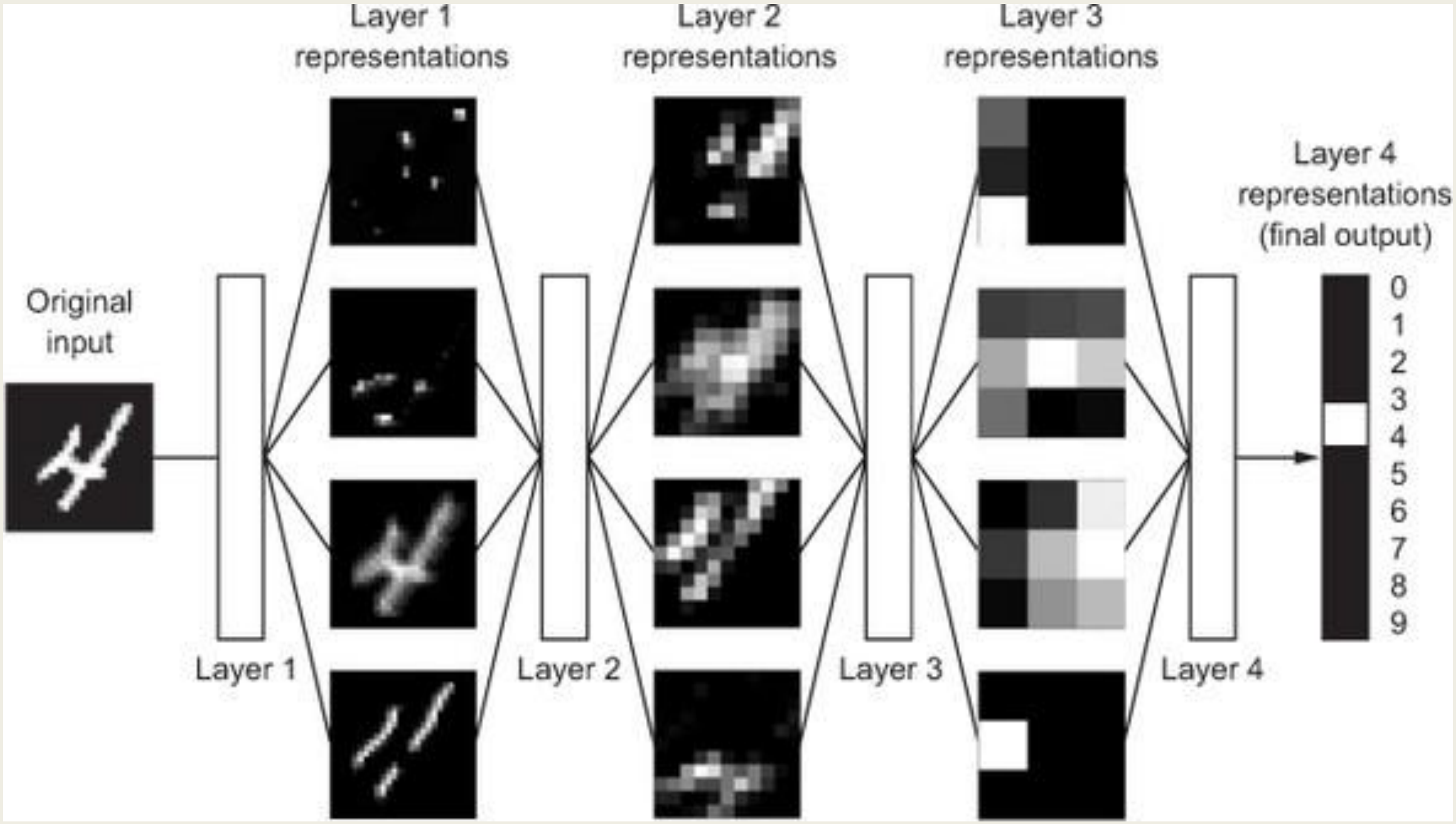
Lessons From Biology: Layers of Neurons



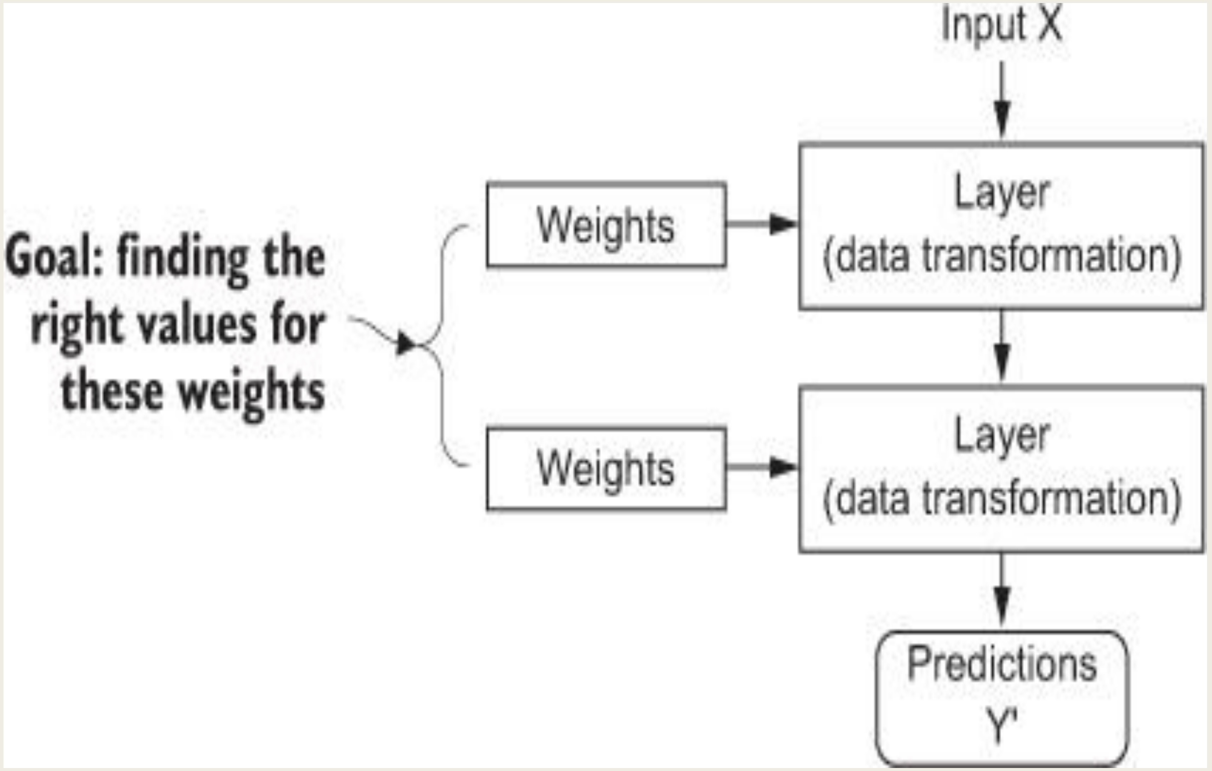
Examples


Layer 1
Layer 2
Layer 3
Layer 4
Original input
0
1
2
3
4
5
6
7
8
9
Final output

Another Example

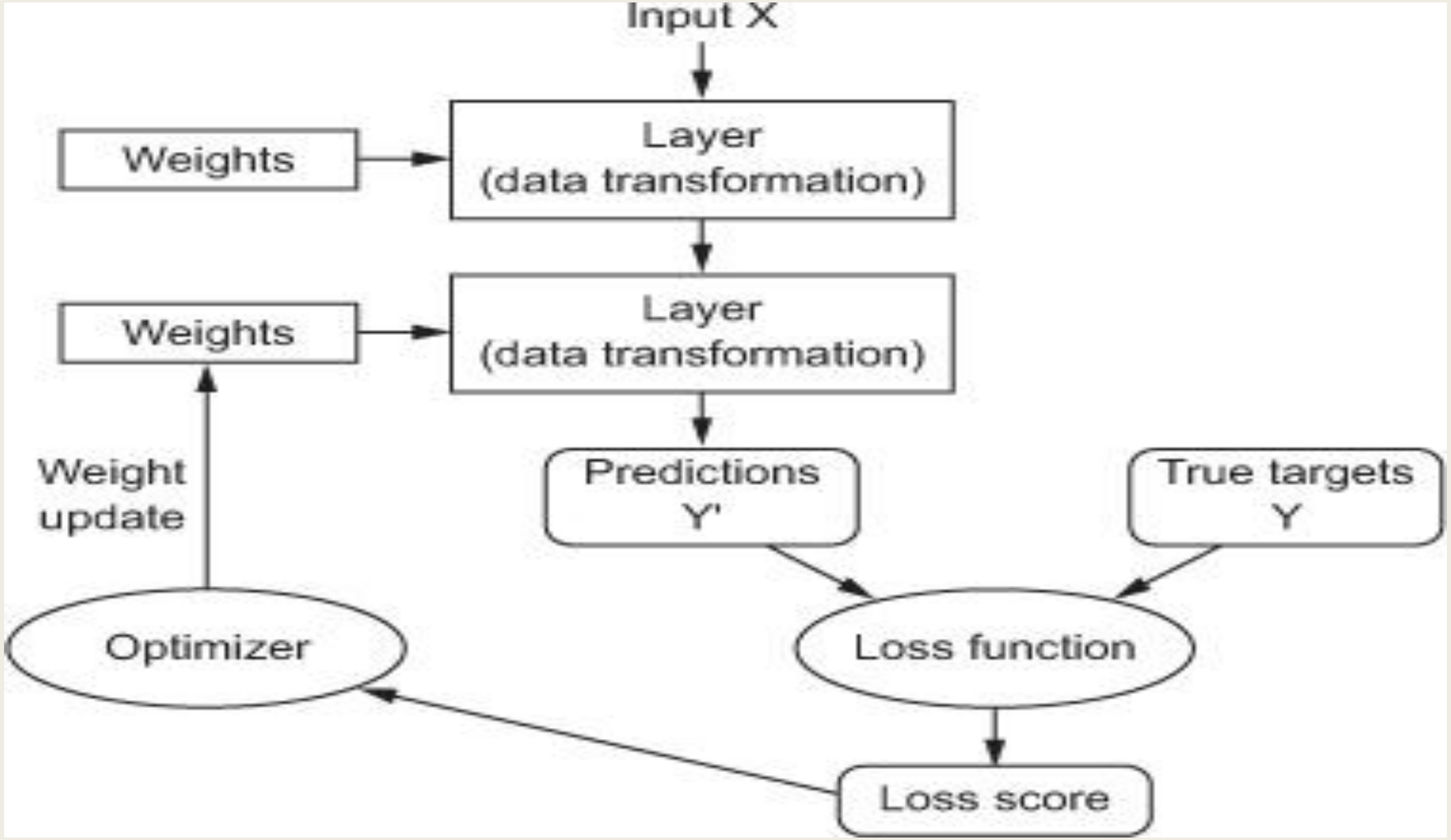


The Weights



Input X
Goal: finding the right values for these weights
Weights
Layer (data transformation)
Weights
Layer (data transformation)
Predictions Y'

The Loss Function: The Learning Process



Important Parameters

To perform training and testing, we need to decide on three important parameters:

- *A loss function*— How the network will be able to measure its performance on the training data, and thus how it will be able to steer itself in the right direction.
- *An optimizer*— The mechanism through which the network will update itself based on the data it sees and its loss function.
- *Metrics to monitor during training and testing*— Here, we'll only care about accuracy (the fraction of the images that were correctly classified).

Training and Deploying the Model:

The deep learning process consists of three steps:

- Training the model using the training portion of the data set
- Test the model using the testing portion of the data set
- Save the trained model
- Deploy the saved model to perform real tasks